

TD2- add-in concepts -Datawarehouse

Pedagogical objective: This document presents the concepts that were introduced during the TD2.

Key concepts to remember

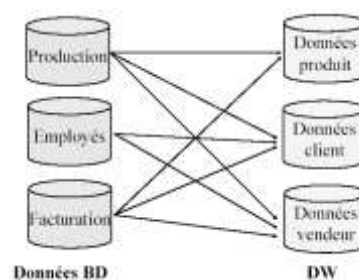
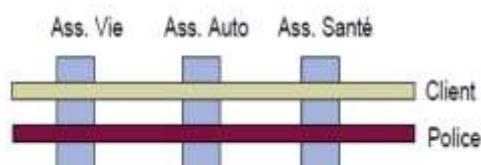
1. The "Datawarehouse" or "DataMart" database

1. The Datawarehouse:

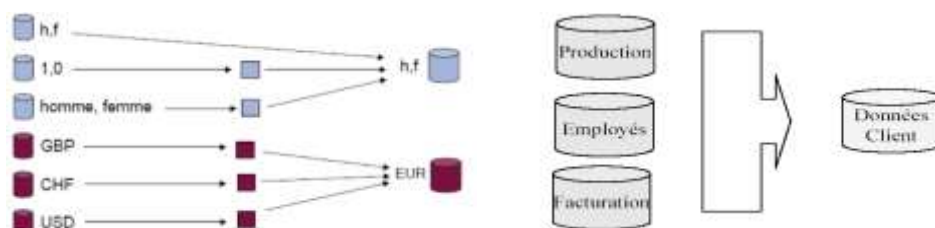
- It is a database that hosts previously selected operational data from various internal and external sources. This data will be organized, processed and stored so that it can be used for decision analysis purposes;
- « *The data warehouse is a copy of transaction data specifically structured for request and analysis.* » (Kimball et Ross, 2011).
- « *A collection of subject-oriented, integrated, non-volatile, and evolving data organized to support a decision support process.*» (Inmon, 2002).
- **The DataMart :**
 - It is a reduced model of a data warehouse because it focuses on a single area of the organization, unlike a ED which covers all the company's business lines.
 - B. Inmon: " *The Datamart is the result of a flow of data from the Datawarehouse. Unlike the datawarehouse, which presents the details of the data for the entire company, it is intended to present the data in a specialized, aggregated and functionally grouped way* ».
 - R.Kimball: " *The Datamart is a subset of the Datawarehouse, made up of tables at the detailed level and at more aggregated levels, making it possible to restore the entire spectrum of a business activity*»

2. Datawarehouse or DataMart data characteristics:

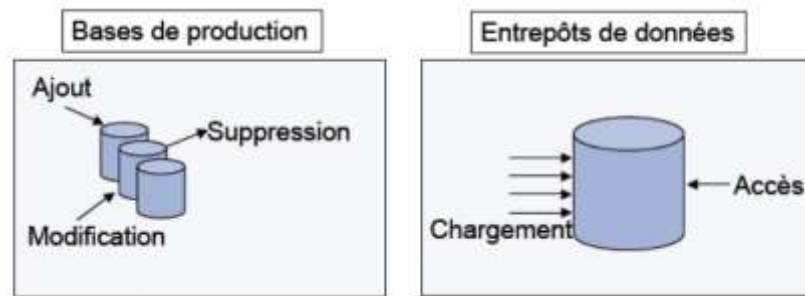
- **The data is subject oriented:** the data collected and to be stored must be business-oriented and therefore sorted by theme (Finance, Sales, Production,...)



- **Integrated data:** As data comes from different sources with different formats, data integration is required, in order to homogenize and streamline it before storing it,



- **Non-volatile data:** the data is not updated, it is not erasable, and is logged (historized)
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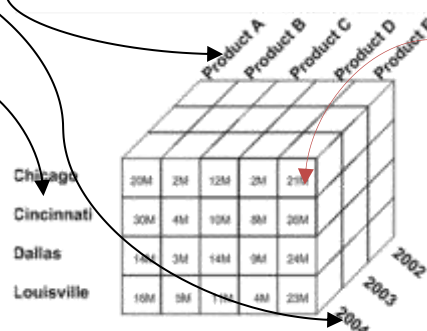
- **Data that evolves over time:** as each piece of data is historical, this makes it possible to analyze its evolution over time (e.g., comparing the sales achievements of one year to another.).)

3. **Differences between a DW database and an operational database :** There is a big difference between the data warehouse and the operational domain (transactional - OLTP):

- Transactional databases (finance, HR, Sales,...) are intended for daily operations with simple queries: Add, update and delete, and consult with statistics: the turnover of a period, by customer or by product.
- The dataWarehouse is intended for business intelligence and data analysis, it allows:
 - Gather and organize information from various sources,
 - Integrate and store data for a business-oriented view (as needed),
 - Find and analyze information quickly and easily for decision support.

4. **Multidimensional modeling:**

- **R.Kimball :** « *Multidimensional modeling consists of considering a subject analyzed as a point in a multidimensional space. The data is organized in such a way as to highlight the topic being analyzed and the different perspectives of the analysis* »



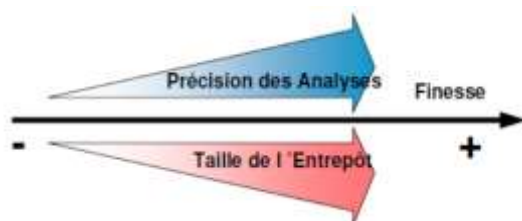
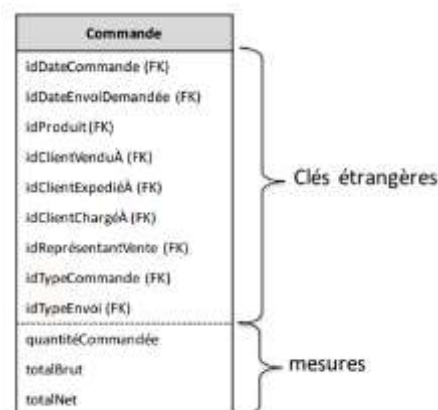
• **Two levels of modeling :**

- **Conceptual:** The aim is to develop the dimensional model of the data warehouse independently of the technical choices,
- **Logical:** This is to describe the dimensional model according to the OLAP technology to be used,

A dimensional model consists of a **table of facts** and related **tables of dimensions**.

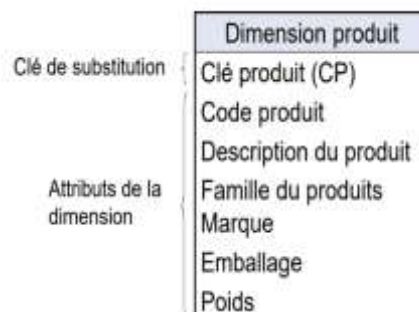


- **The concept of fact:** It is a table of facts that contains the measures and foreign keys, which are the primary keys of the dimension tables related to it. The fact table is the largest table.
- **The concept of grain of fact:** Refers to the meaning of a line in the table of facts. "Grain is the level of detail associated with the measurements in the fact table" (Kimball and Ross, 2011). It is advisable to opt for the finest grain of the information to be analyzed so that it cannot be subdivided, but be careful :



- The grain depends on the availability of the source data. *Example - hypermarket: a line of the receipt or the total of the receipt or the stock at the end of the day?*
 - Recommendation: Use the most detailed information whenever possible. (Opt for a row in the fact table that records sales by date and time instead of a day, week, or month rollup)

- The concept of dimension:** A dimension models a perspective of the analysis, it is a table made up of a set of attributes used to evaluate and analyze a fact. It is generally consisting of: an artificial key, a natural key and attributes. The analytical power of the DE is proportional to the richness and quality of the dimensional attributes.



- **The concept of dimensional hierarchy:** A hierarchy level is defined at the level of dimension tables. A hierarchy organizes the attributes of a dimension according to a "is finer" relationship in accordance with the hierarchy relationship that exists between attributes. The hierarchy allows for the analytical power of the facts in relation to the attributes of the dimensions, in other words, it allows the definition of paths for the access and analysis of the facts :
 - **Temps** : année→mois→semaine→jour→heure
 - **Produit** : famille→catégorie→marque→produit
 - **Lieu** : pays→province→région→ville→code postal

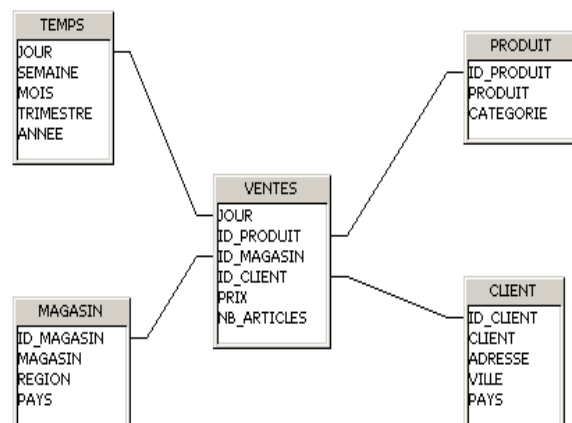
- The time dimension:** This is a center-dimensional table in a dimensional model. It is necessary because the analysis is temporal (evolution of sales, number of customers, absenteeism rate,...)
- Depending on the context and the availability of the source data, the finest level of detail should be expected and all values should be expected even if they can be inferred.

Dimension: Date
idDate (PK)
date
jourDeSemaine
jourDuMois
jourDeAnnée
jourDansMoisFiscal
jourDansAnnéeFscale
congéférié
jourDeTravail
semaineDuMois

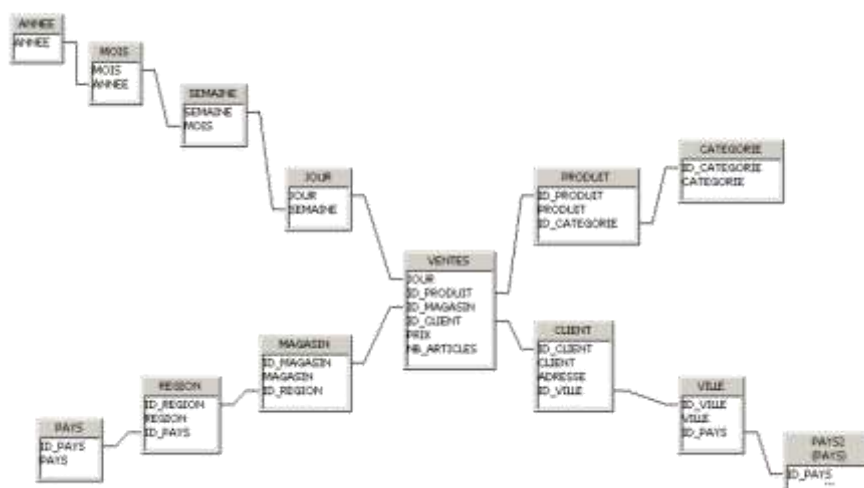


1. The star dimensional model:

1. This model has a star shape. There is a de facto table in the center surrounded by the tables of dimensions associated with it. It is a highly denormalized representation that ensures a high level of query performance even on large volumes of data,
2. This modeling, as opposed to the standardized 3NF schemas, makes it possible to respond to two main characteristics of decision-making systems: the performance and simplicity of the queries.
3. There are two variants of this model:
 - Snowflake model
 - Constellation model



- **The snowflake dimensional model:** This is a star model, except that it avoids redundancy by normalizing dimension tables. This approach reduces the performance of the data warehouse (joins).



- **The Constellation dimensional model:** This is a diagram that merges several star diagrams. It contains several fact tables that have tables of common dimensions (represents a Datawarehouse).

